

Open Source CFD

Tutorial #4: 3D Y-Pipe

Incompressible Steady Flow: 3D Y-Pipe with simpleFOAM

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Learning Objectives:

- Create a complex 3D geometry with sketch and additive pipe with FreeCAD
- Generate high quality STL surface
- Mesh a 3D geometry with `snappyHexMesh`
- Setup a case with multiple inlets
- Analyze flow and pressure drop
- Visualize 3D results with `paraview`

This tutorial introduces the fundamental workflow of Computational Fluid Dynamics (CFD) for a 3D geometry. We simulate the flow in a Y-shaped pipe junction. You will create the 3D geometry with FreeCAD, generate a high-quality STL file, mesh the geometry, run the steady simulation and, finally, post-process the results.

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1 Introduction

The Y-pipe consists of:

- Main pipe: 250 mm long, 100 mm diameter
- Secondary branch: 500 mm long branch at 90° , 100 mm diameter
- Inlet branch: 250 mm long, 50 mm diameter

An scheme of the Y-pipe is shown in Figure 1

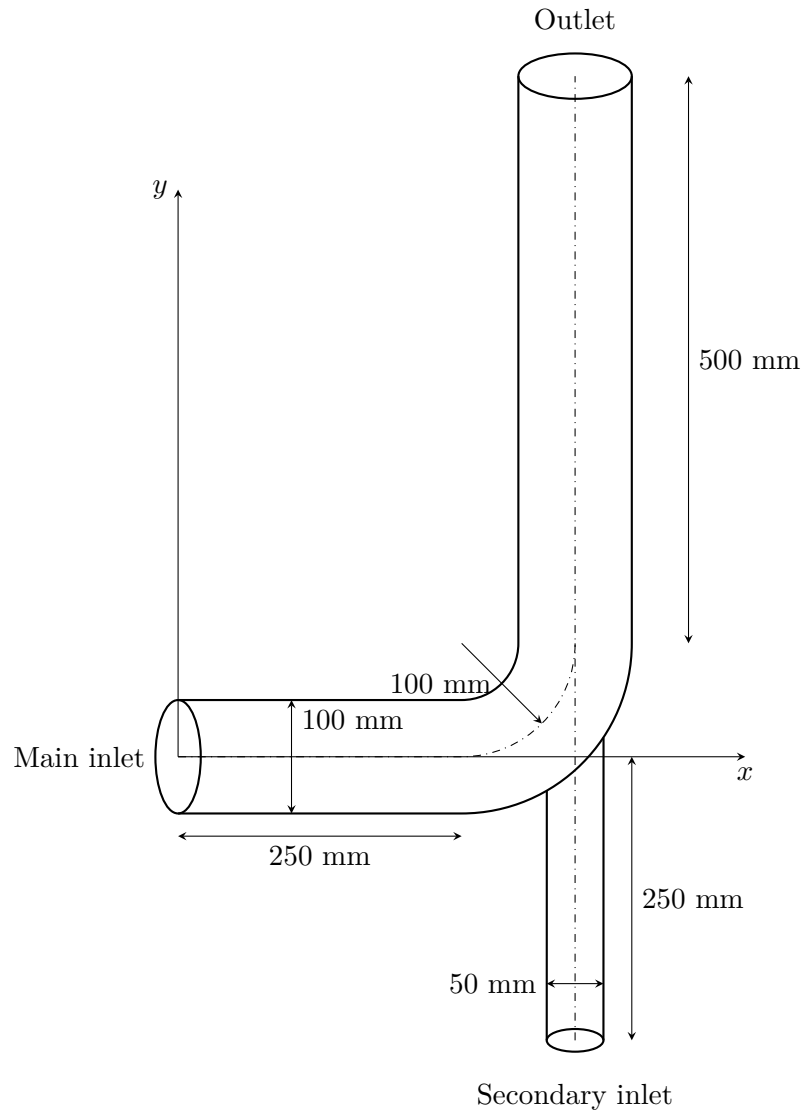


Figure 1: Y-pipe geometry schematic

2 Generation of the geometry

First, open a new **Part design** and an sketch in the XY plane. Draw the center line of the main pipe and the branch. It is composed of an horizontal line of 250 mm long, an arc of radius 100 mm and a vertical line 500 mm long. Don't worry about the dimensions and tangent lines when drawing the sketch, they can define later. When the sketch is completely defined, it becomes green, and it can be closed.

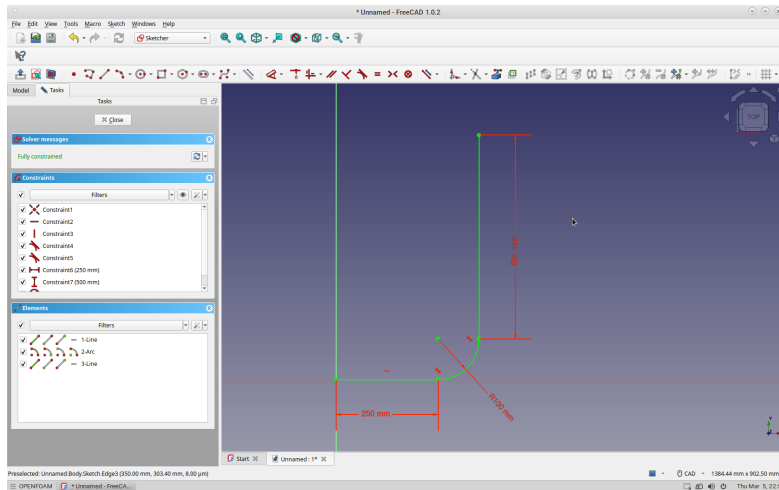


Figure 2: First sketch.

Draw another sketch, with just a circle of radius 50 mm, in the YZ plane

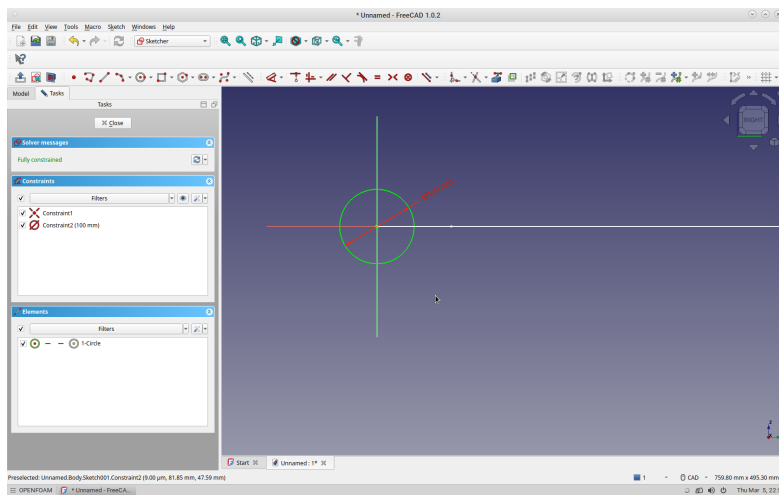


Figure 3: Sketch for 100 mm circle in YZ plane

Select both sketches and make an **Additive pipe**. The result should be like in Figure 4.

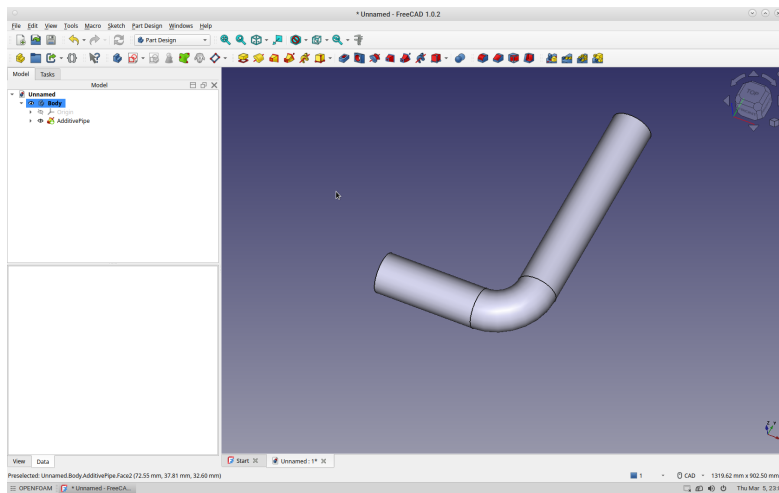


Figure 4: First main pipe

Let's now make the secondary branch. Create another sketch, but in the **outlet** face, with a 50 mm diameter circle, with center coincident to x axis, and 350 mm away from z axis (see Figure 5)

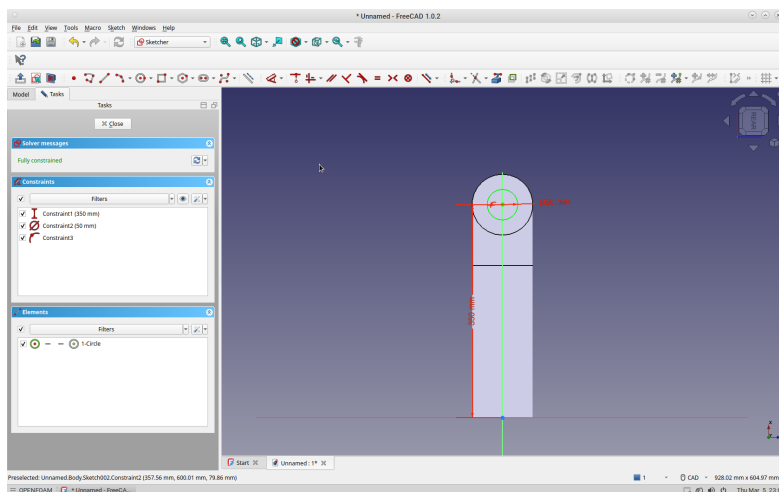


Figure 5: Circle in outlet face.

Make a Pad of 800 mm with this circle with reversed direction, and the geometry is finished. **Don't forget to save the work!** Save it with the name **Ypipe** in a **Tutorial4** folder.

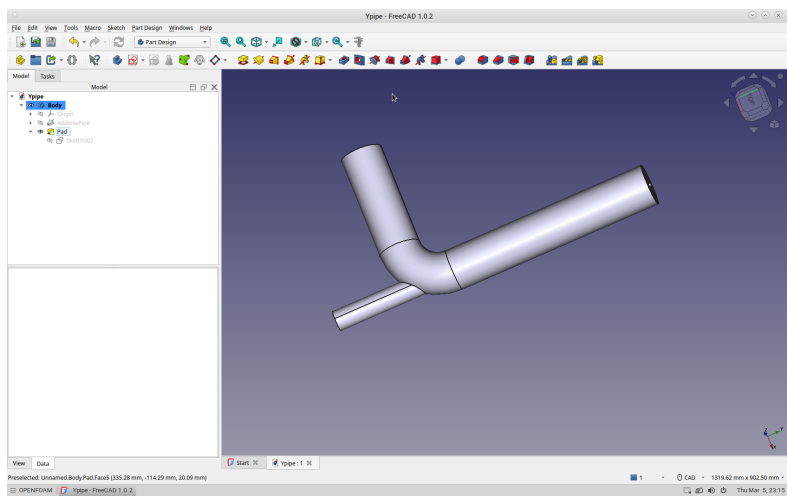


Figure 6: Y-pipe geometry.